National 5 Engineering Science

Assignment: mark it yourself



R Stewart - November 2026

Remember

This pack is for practice **before** your assignment. The real assignment is done on your own, with no help.

Do not copy answers from these scripts. Each year has a new task.

Read this on the computer. Write your answers on the printed **worksheet**.

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1 How to use this pack

Three real assignments are in this pack. They come from Qualifications Scotland's Understanding Standards website. A marker has already marked them.

Your job: mark them yourself first. Then compare your marks with the marker's.

Steps

Step 1: **Read** where the marks are lost, in part 2.

Step 2: **Skim** the 2024 task in part 3. It says what each candidate was asked to do.

Step 3: **Read** one script: parts 5 to 7. Use the marking instructions in part 4 to mark it.

Step 4: **Fill in** that script's grid on your worksheet.

Step 5: **Complete** the focus tasks on your worksheet.

Step 6: **Check** your marks against the answers in part 8.

Why marking works

When you mark, you see what the marker looks for. You learn the standard. Most marks are lost by leaving out one small thing that the marking instructions ask for.

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2 Where the marks are lost

The assignment is worth 50 marks. The table runs from the most marks lost to the fewest. It uses the 2024 results for all 2,000 pupils in Scotland who sat this task.

Task	What it asks for	Max	Scotland average	Marks lost
2c	Test results and amendments	6	3.0	3.0
3a(ii)	Sub-system diagram	6	3.1	2.9
1b	Pneumatic circuit	5	3.6	1.4
3c	Test plan: Vout	3	1.6	1.4
2a	Flowchart and circuit	5	3.9	1.1
5c	Modify and justify	2	1.0	1.0
3a(i)	System diagram: input and outputs	2	1.0	1.0
4a	Evaluate the gear design	3	2.0	1.0
2d	Amendments carried out	2	1.3	0.7
4d	Velocity ratio	2	1.4	0.6
5a	Logic diagram	2	1.5	0.5
4b	Design a gear train	2	1.5	0.5
1a	Actuator and justification	2	1.5	0.5
3b	Input sensing circuit	3	2.5	0.5
2b	High-level code	1	0.5	0.5
4c	Simulate the gear train	2	1.6	0.4
5b	Simulate the logic	2	1.8	0.2

Source: Centre Data 2024, all centres in Scotland.

Biggest losses

Testing (2c) and the sub-system diagram (3a(ii)) lose the most marks. In both, every mark needs one exact thing. Check the marking instructions line by line.

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3 The 2024 task: toy shop

These are the pages each candidate was given: five tasks for a toy shop. **Skim** them now. Come back to a task when you mark it.

National 5 Engineering Science Assignment 2024, assessment task, pages 12 to 23. © Qualifications Scotland (SQA).

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Toy shop

A team of engineers is involved in developing a range of systems for a toy shop.

The tasks include developing proposals for the following:

Task 1 — model railway

Task 2 — interactive teddy bear

Task 3 — shop front night setting

Task 4 — rotating display stand

Task 5 — toy safe

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Task 1 — model railway

A model railway display in the toy shop lets trains travel around a series of tracks.



It needs a pneumatic circuit to switch the rails between two sets of tracks, then return them automatically after the train has passed.

The pneumatic circuit must meet the following specification:

i When a push button actuator on a 3/2 valve is pressed, a 5/2 valve will cause the piston in a double-acting cylinder to outstroke slowly and smoothly.
ii An actuator on a second 3/2 valve will detect when a train has passed.
iii After a pneumatic time delay, the piston will instroke slowly and smoothly.

The table below shows possible actuators that could detect when a train has passed.

Type	Actuator
Electrical	solenoid
Mechanical	roller, roller trip, plunger
Pneumatic	diaphragm
Manual	lever, push button

1a Choose a suitable actuator from the table above, to sense when a train has passed.

Complete **worksheet 1a**, justifying your choice.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(2 marks)

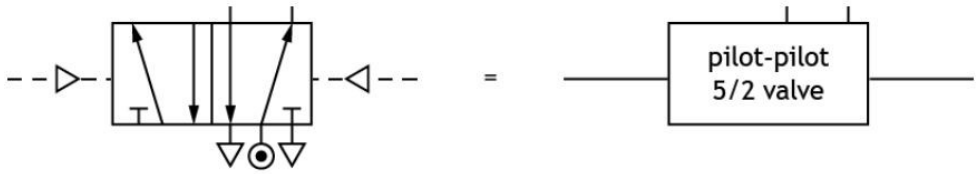
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Task 1 — model railway (continued)

1b Complete the design for the pneumatic circuit on **worksheet 1b**, to meet the specification, using your selected actuator from **task 1a**.

You must identify the name of each component, valve, and actuator, and indicate the direction of the piston outstroke. Show connections between all components and valves.

You can simplify components. For example, you could draw a 5/2 valve as shown below.



Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(5 marks)

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Task 2 — interactive teddy bear

The toy shop owner wants an interactive teddy bear to entertain customers. It will have a motorised bow-tie with flashing LEDs.

An electronic engineer has designed a possible microcontroller-based solution.

The pin numbers used to connect an electronic circuit to the microcontroller are shown in the table below.

Input	Pin	Output
	7	motor
	5	LEDs
test switch	2	
movement sensor	0	

2a Simulate the flowchart sequence and electronic circuit **integrated together,** as shown on the following page. You can use a different sized microcontroller but the pin numbers must match the connections given.

Note: for test purposes, the movement sensor is replaced with a push-to-make switch.

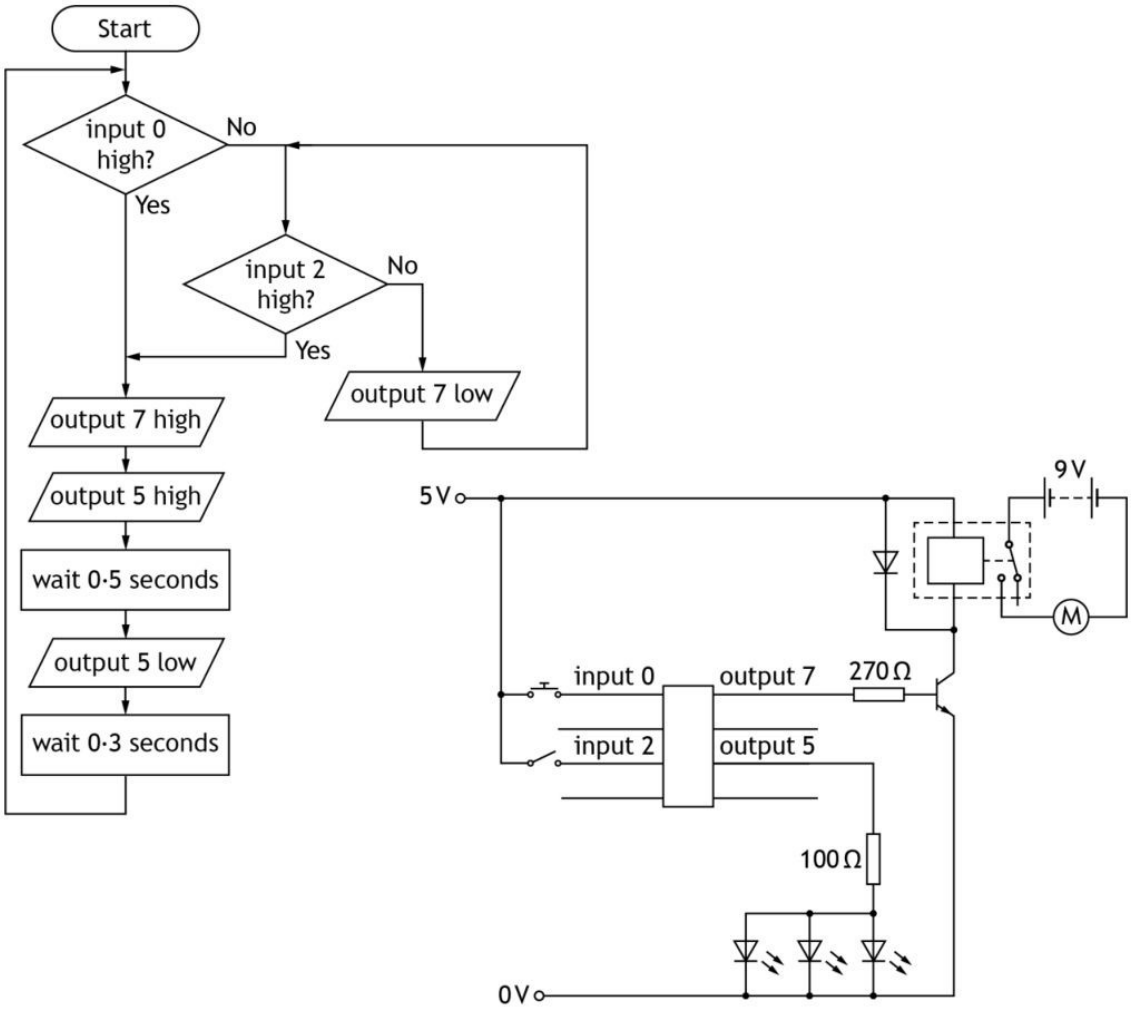
Present your evidence on A4 single-sided pages, with the task number clearly labelled. Screenshots or images must be clear and easy to read.

(5 marks)

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Task 2 — interactive teddy bear (continued)

Microcontroller-based solution



2b Produce high-level microcontroller code to fully match the flowchart from **task 2a**.

Present your evidence on A4 single-sided pages, with the task number clearly labelled. Screenshots or images must be clear and easy to read.

(1 mark)

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Task 2 — interactive teddy bear (continued)

The microcontroller must meet the following specification:

i Whenever the test switch (input 2) is activated, the motor will rotate, and the LEDs will flash on and off continuously.
ii Whenever the movement sensor (input 0) detects a customer, the motor will rotate, and the LEDs will flash on and off continuously.
iii When the movement sensor no longer detects a customer, the motor and the LEDs will switch off.

Errors were found in both the flowchart and electronic circuit during testing.

2c Complete the testing table on **worksheet 2c**, by carrying out the planned tests and describing each initial test result.

Describe your planned amendments to the flowchart and/or circuit, so that it meets the specification and make the amendments.

Carry out each test again, describing the result of your retest, before moving onto the next test.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(6 marks)

2d Your final amended flowchart and electronic circuit should now meet the specification given at the start of **task 2c**.

Present your evidence of your amended flowchart and electronic circuit integrated together, on A4 single-sided pages, with the task number clearly labelled. Screenshots or images must be clear and easy to read.

(2 marks)

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Task 3 — shop front night setting

A microcontroller-based circuit is needed to operate a light above the shop's sign and a motorised blind in the shop window. The circuit must meet the following specification:

i A sensor will detect the outside light level.
ii When it is dark, the microcontroller will switch on the lamp and turn the motor to fully close the blind.
iii When it is light, the microcontroller will then switch off the lamp and turn the motor to fully open the blind.
iv Two limit switches will send feedback signals to the microcontroller. The microcontroller will stop the motor when either the blind is fully open or fully closed.

3a(i) Complete the system diagram on **worksheet 3a(i)** to meet the specification.

You must clearly show all external inputs and outputs.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(2 marks)

(ii) Complete the sub-system diagram on **worksheet 3a(ii)** to meet the specification.

You must clearly show all external inputs and outputs, all sub-systems, the system boundary and the interactions between sub-systems.

Note: each output device requires a driver.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(6 marks)

A dark sensor is needed to detect the outside light level and send a signal to the microcontroller.

3b Complete the circuit diagram on **worksheet 3b,** by designing a suitable input sensing circuit that will produce a high signal (V_{out}) when it is dark. Component values are not required.

Note: you must not construct or use simulation software to complete this part of the task.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(3 marks)

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Task 3 — shop front night setting (continued)

3c Using the input sensor circuit you designed in **task 3b,** complete the test plan on **worksheet 3c** by describing:

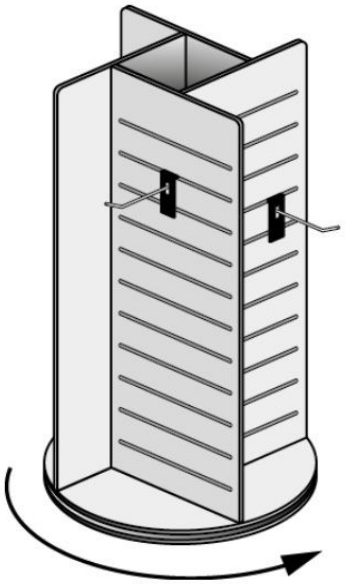
- the expected result from the planned test given in terms of V_{out}
- a second planned test and the expected result in terms of V_{out}.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(3 marks)

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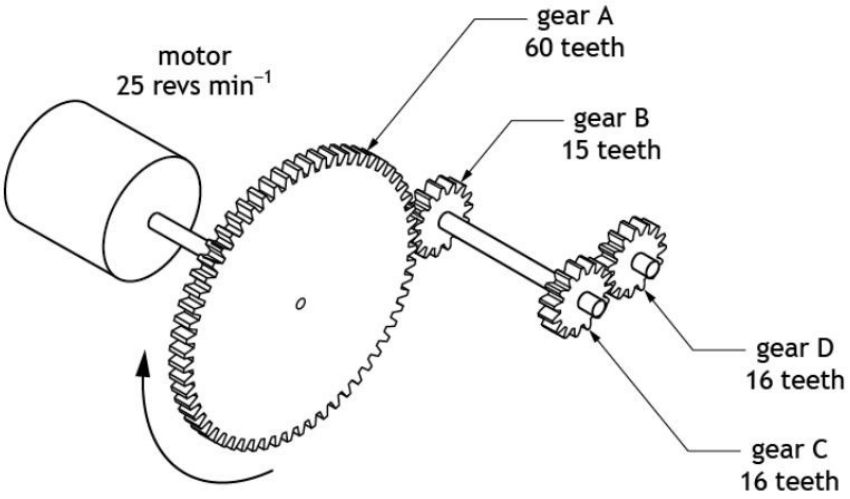
Task 4 — rotating display stand



A rotating display stand needs a motorised gear train that will meet the following specification:

i It must fit into a compact space inside the rotating display stand.
ii The output gear D must turn in the same direction as the input gear A.
iii It must reduce the speed of the output by at least a factor of 6 but not greater than a factor of 9.

An initial design for the gear train is shown below, with gear sizes and input speed.



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Task 4 — rotating display stand (continued)

4a Evaluate the initial gear train design against the specification and complete the table on **worksheet 4a**. You should:

- state if each specification point is met
- justify your decision

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(3 marks)

The display stand speed is found to be unsuitable.

4b Design a gear train on **worksheet 4b,** based on the gear sizes available to you. This needs to give a speed reduction of at least a factor of 12, but not greater than a factor of 20. Label the input gear and include all gear sizes.

Note: you must not construct or use simulation software to complete this part of the task.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(2 marks)

4c Simulate or construct your design from **task 4b**. You must include an input component to allow for testing and label all gear sizes.

Present your evidence on A4 single-sided pages, with the task number clearly labelled. Screenshots or images must be clear and easy to read.

(2 marks)

4d Complete the test result table on **worksheet 4d** by:

- entering the results from **task 4c** that you observed during simulation or after construction
- calculating the actual velocity ratio, using your recorded values for the input and output speeds

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(2 marks)

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Task 5 — toy safe

The shop owner wants to run a competition where customers try to solve the code to open a toy safe and win a prize.



5a Complete the logic diagram on **worksheet 5a** to unlock the safe by sending a high (1) signal to output Z when:

input A is high (1) **and** input B is low (0) **and** input C is high (1)

Note: you must not construct or use simulation software to complete this part of the task.

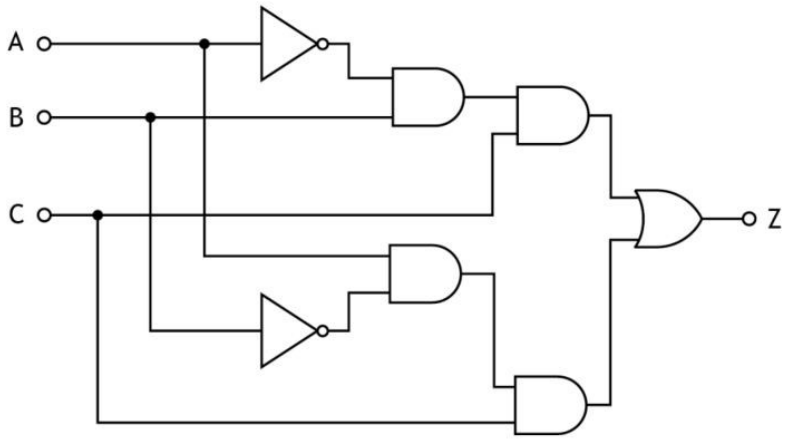
Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(2 marks)

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Task 5 — toy safe (continued)

Another code for the toy safe was designed using the logic diagram shown below.



5b Simulate or construct the logic diagram shown above. You must include suitable input and output devices to allow for potential testing. If constructed, you must clearly label the integrated circuit (IC) numbers.

Present your evidence on A4 single-sided pages, with the task number clearly labelled. Screenshots or images must be clear and easy to read.

(2 marks)

5c Using **worksheet 5c,** describe a suitable modification to the logic circuit in **task 5b** that would make the code for the toy safe harder to solve and justify your reason.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(2 marks)

[END OF ASSIGNMENT]

End of the task.

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4 The marking instructions

This is what the marker uses. Each bullet with "(1 mark)" is one mark. **Mark** each script against these pages.

National 5 Engineering Science Assignment 2024, finalised marking instructions. © Qualifications Scotland (SQA).

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General marking principles

This information is provided to help you understand the general principles that must be applied when marking candidate responses in this assignment. These principles must be read in conjunction with the detailed/specific marking instructions, which identify the key features required in candidate responses.

a Marks for each candidate response must always be assigned in line with general marking principles and the specific marking instructions for this assessment.

b Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.

c If a specific candidate response is not covered by either the general marking principles or detailed marking instructions, you must seek guidance from your team leader.

02

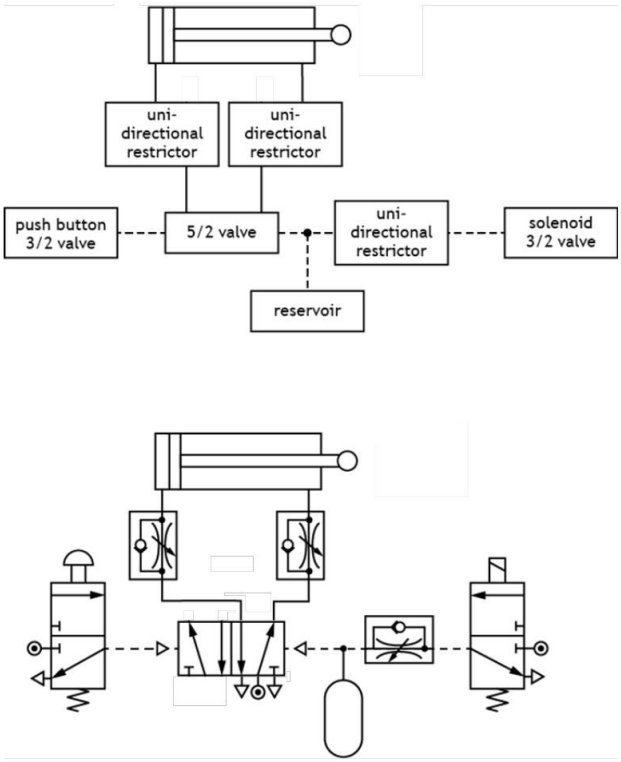
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Detailed marking instructions

Task			Expected answer(s)	Max mark	Additional guidance
1	a		Diaphragm This actuator would act as a contactless sensor as the train passes, providing a longer lifespan for the sensor. Or Solenoid This actuator can be combined with a microswitch which senses model trains passing. Or Roller trip This actuator will detect trains in one direction of travel.	2	• Appropriate actuator selected: Any electrical, mechanical or pneumatic actuator. **(1 mark)** • An appropriate justification for selected actuator in context. **(1 mark)** Do not accept a manual actuator. Apply FTE for justification.

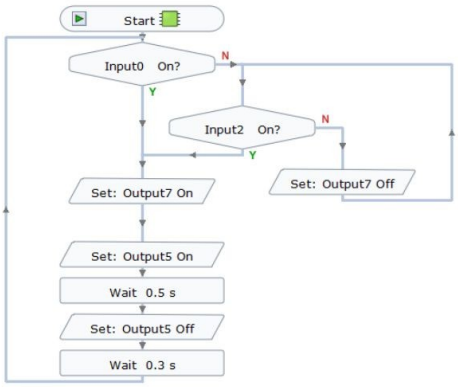
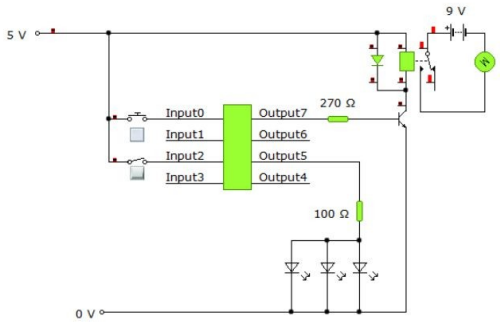
03

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Task			Expected answer(s)	Max mark	Additional guidance
1	b			5	• Double-acting cylinder to 5/2 valve both pipes (with direction of outstroke indicated correctly if required). **(1 mark)** • 2 x uni-directional restrictor used to slow instroke & outstroke of cylinder. **(1 mark)** • 2 x 3/2 value to cause outstroke and instroke piped to 5/2 valve. **(1 mark)** • Push button for outstroking **and** selected actuator, from task 1a, for instroking. **(1 mark)** • Reservoir after instroking 3/2 valve (with optional uni-directional restrictor). **(1 mark)** Block diagram, circuit diagram (or a hybrid) or constructed/simulated. Components identified or implied by name or use of components (symbols). Connections between components and valves must be shown. Ignore line types. Port to port piping is not required. Ignore orientation of uni-directional restrictor symbols. Allow FTE for selected actuator (if no response for task 1a, accept any from given table).

04

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Task			Expected answer(s)	Max mark	Additional guidance
2	a		 	5	**Flowchart** • All stages with correct symbols (to the software) and connections. **(1 mark)** • All correct pin numbers, pin states, decisions and time delays (to the software). **(1 mark)** **Electronic circuit** • All correct components selected. **(1 mark)** • Correct wiring, microcontroller pin connections, component orientations and values. **(1 mark)** **Integration** • Of flowchart and electronic circuit. **(1 mark)** Note: Allow use of alternative-sized microcontrollers.

05

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Task			Expected answer(s)	Max mark	Additional guidance
2	b		main: label0: if Input0 = 1 then label1 label2: if Input2 = 1 then label1 low 7 goto label2 label1: high 7 high 5 pause 500 low 5 pause 300 goto label0 goto label1	1	• Code to fully match the flowchart, including pin numbers, in task 2**(a). (1 mark)** Accept manually written/automatically generated. Any high-level language acceptable.

06

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<table>
<tr><th colspan="3">Task</th><th>Expected answer(s)</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>2</td><td>c</td><td></td><td>
<table>
<tr><th>Test 1:
Initial test result</th><th>Planned amendment to circuit/flowchart</th><th>Re-test result</th></tr>
<tr><td>When the test switch is turned on…
LEDs flash and the motor did not turn.

When the test switch is turned off…
the LEDs turn off and the motor slows to a stop.</td><td>Rotate the diode 180^0.

No amendment.</td><td>When the test switch is pressed…
the motor turns on and the LEDs flash

When the test switch is turned off…
the LEDs turn off and the motor slows to a stop.</td></tr>
<tr><th>Test 2:
Initial test result</th><th>Planned amendment to circuit/flowchart</th><th>Re-test result</th></tr>
<tr><td>When the movement sensor is pressed…
the motor and LEDs do not turn on.

When the movement sensor is released…
the motor and LED remained off.</td><td>Alter loop back from output 7 off to the start of flowchart.

No amendment.</td><td>When the movement sensor is pressed…
the motor turns and LEDs flash.

When the movement sensor is released…
the motor and LEDs turn off.</td></tr>
</table>
</td><td>6</td><td>Mark Initial Test Result and Planned Amendment responses based on 2(a) evidence.

Mark Re-test result based on 2(d) evidence.

If no evidence for 2(a)/2(d), then mark as below.

• Description of LED(s) flashing and the motor did not turn. (1 mark)

• Description of changing the diode orientation. (1 mark)

• Description of re-test result: motor starts to turn and the LED(s) flash. (1 mark)

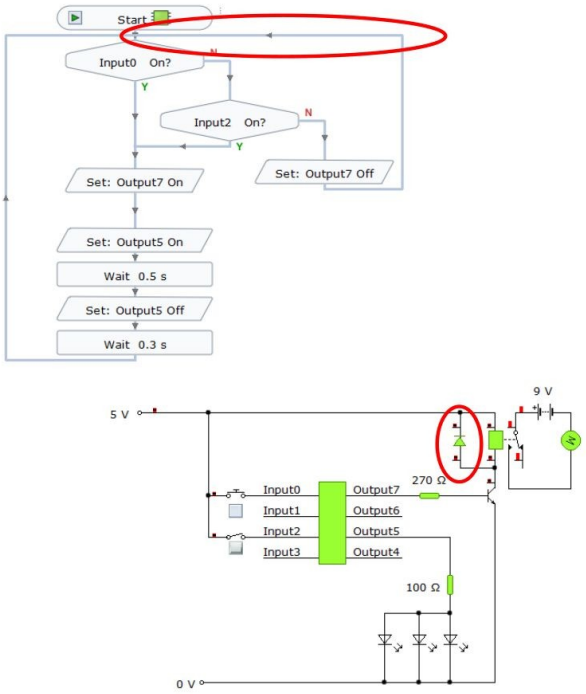
Apply FTE from Test 1 responses.

• Description of LED off and motor not turning. (1 mark)

• Description of connection from output 7 off returning to start. (1 mark)

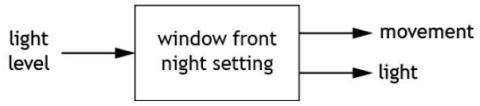
• Description of re-test result: motor starts and LED(s) flash and (when movement sensor released) LED(s) switch off and motor slows to a stop. (1 mark)</td></tr>
</table>

07

Task			Expected answer(s)	Max mark	Additional guidance
2	d		 	2	• Candidate's planned amendment 1 carried out: correction of protective diode orientation. **(1 mark)** • Candidate's planned amendment 2 carried out: Connection from output 7 off returning to start. **(1 mark)**

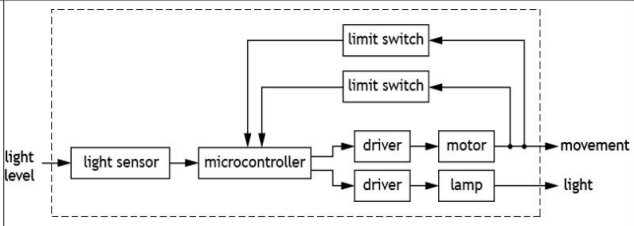
08

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Task			Expected answer(s)	Max mark	Additional guidance
3	a	i		2	• Light/dark level input identified (implied). **(1 mark)** • Light **and** movement (implied) outputs identified. **(1 mark)** Accept light/daylight/darkness on its own as input. Do not accept light sensor. Do not accept components/devices. Ignore extra boxes or words.

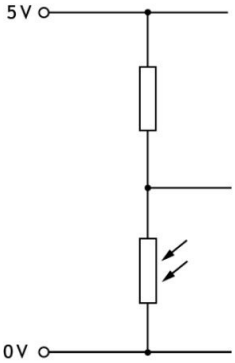
09

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Task			Expected answer(s)	Max mark	Additional guidance
3	a	ii		6	• System boundary around sub-systems only and boxes around each sub-system. **(1 mark)** • Motor in output position. **(1 mark)** • Lamp in output position. **(1 mark)** • One driver per output device individually connected to microcontroller. **(1 mark)** • 2 (limit) switches connected individually to the microcontroller. **(1 mark)** • Feedback loop(s) from after motor via (limit) switch(es) into microcontroller with both arrowheads. **(1 mark)** Accept 'motorised blind'. Do not accept 'light', 'bulb' or 'LED' instead of 'lamp'. Do not accept an action (eg 'lamp on'). Ignore extra sub-systems. If no arrows shown between sub-systems, assume left to right.

10

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Task			Expected answer(s)	Max mark	Additional guidance
3	b			3	• Correct symbol for LDR. **(1 mark)** • Correct symbol for fixed/variable resistor. **(1 mark)** • Correct position of components connected to create a dark sensor. **(1 mark)** Values of components are not required. Do not accept simulation/construction evidence. Ignore additional circuitry.

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<table>
<tr><th colspan="3">Task</th><th>Expected answer(s)</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>3</td><td>c</td><td></td><td>
<table>
<tr><th>Planned test</th><th>Expected result</th></tr>
<tr><td>Test 1

Reduce the light level.</td><td>V_{out} will increase.</td></tr>
</table>
<table>
<tr><td>Test 2

Increase the light level.</td><td>V_{out} will decrease.</td></tr>
</table>
</td><td>3</td><td>
• V_{out} increases. (1 mark)

• Test for rising light level. (1 mark)

• Expected result from given test in terms of V_{out}. (1 mark)

Apply FTE for expected result from Planned test 2.

If no planned test 2 given, then no mark can be awarded for its unless expected result statement includes a description of the planned test.
</td></tr>
</table>

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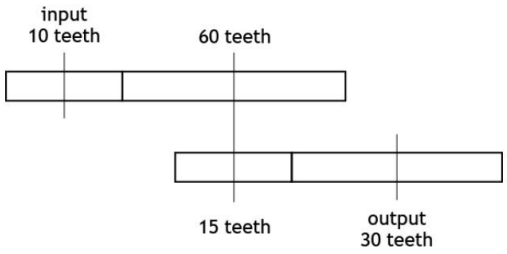
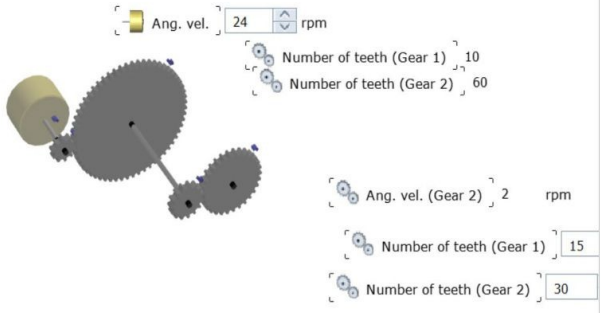
<table>
<tr><th colspan="3">Task</th><th>Expected answer(s)</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>4</td><td>a</td><td></td><td>
<table>
<tr><th>Specification point</th><th>Met? Yes/No</th><th>Justification</th></tr>
<tr><td>i.</td><td>Yes</td><td>As the gear system is a compound gear train.</td></tr>
<tr><td>ii.</td><td>Yes</td><td>The input gear and output gear both turn in same direction.</td></tr>
<tr><td>iii.</td><td>No</td><td>The speed of the output gear would be increased.</td></tr>
</table>
</td><td>3</td><td>
• State that specification (i) was met (Y)/ not met (N) and description referring to compound gear train/number of gears/ size of gears. (1 mark)

• State that specification (ii) was met (Y) and refers to direction of rotation/acting as an idler. (1 mark)

• State that specification (iii) was not met (N) and refers to increase in output speed (from input drive). (1 mark)

Mark can be awarded for specification (iii) for calculation-based justification which must compare like for like (eg velocity ratio to velocity ratio, rather than velocity ratio to gear ratio).
</td></tr>
</table>

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Task			Expected answer(s)	Max mark	Additional guidance
4	b			2	• Simple/compound gear train/worm & wheel **and** all gear sizes. **(1 mark)** • Correct speed reduction of at least a factor of 12 but no more than 20 **and** labelling input or implied (eg through inclusion of motor or calculations or output labelled). **(1 mark)** Do not accept chain or belt drive. Do not accept simulation/construction evidence.
	c			2	• Simulated or constructed of mechanism type shown in 4(b), with input identified. **(1 mark)** • All gear sizes shown and matching the design in 4(b) or evidence of velocity ratio (eg graph) that proves teeth numbers. **(1 mark)**

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<table>
<tr><th colspan="3">Task</th><th>Expected answer(s)</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>4</td><td>d</td><td></td><td>
<table>
<tr><th>Input speed</th><th>Output speed</th><th>Required VR</th><th>Actual Velocity Ratio</th></tr>
<tr><td>24 revs min^{-1}
or
24 turns</td><td>2 revs min^{-1}
or
2 turns</td><td>Between 12:1 and 20:1</td><td>12:1</td></tr>
</table>
</td><td>2</td><td>
• Input and output speed/turns from given solution in 4(c). (1 mark)

• Correct velocity ratio for input and output speeds given in table (expressed as ratio, number or fraction). (1 mark)

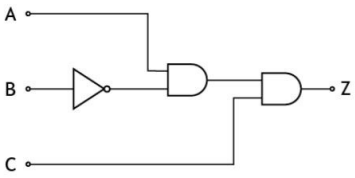
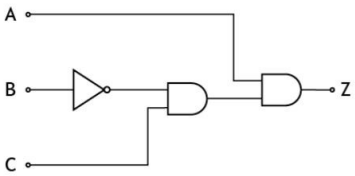
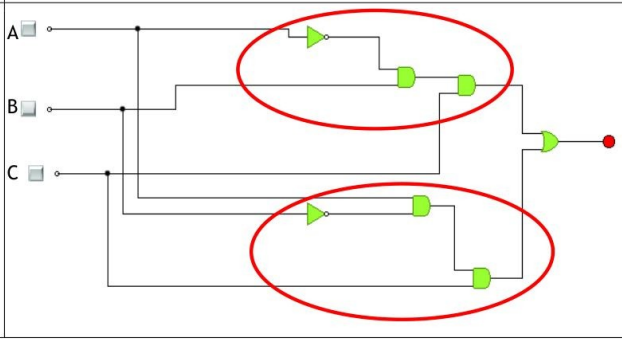
Units not required.

If no sizes/speed indicated in task 4(c) then use values from 4(b) to determine VR.

No evidence of VR (through teeth numbers, graph or velocities) or teeth numbers indicated on either 4(b) or 4(c) then award 0 marks.

No evidence of simulation/construction in 4(c). (1 mark maximum for velocity ratio, if correct)
</td></tr>
</table>

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Task			Expected answer(s)	Max mark	Additional guidance
5	a		 OR 	2	• NOT gate connected to B. **(1 mark)** • AND gate(s) connected to inputs A, B & C **and** output Z. **(1 mark)** Accept alternative correct circuit including three input AND gate. Do not accept simulation/construction evidence.
	b			2	Maximum 1 mark if OR gate not included. • Mark for $\overline{A}$.B.C **(1 mark)** • Mark for A.$\overline{B}$.C **(1 mark)**

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Task			Expected answer(s)	Max mark	Additional guidance
5	c		**Modification**: Increase the number of inputs to the circuit. OR Remove the lower part of circuit to only allow one combination of A/B/C. **Justification**: Increase the number of possible combinations for the toy safe code. OR Increases the chance of getting the code incorrect.	2	• Appropriate suggested modification to the logic circuit. **(1 mark)** • Appropriate justification of suggested modification to the logic circuit. **(1 mark)** Do not accept 'to make it harder to solve' on its own as justification. Do not accept 'replace OR gate with AND gate'. (input combination impossible) The modification must be to the logic circuit rather than alternative solutions. Apply FTE for justification.

[END OF MARKING INSTRUCTIONS]

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End of the marking instructions.

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5 Script 1

Candidate 1's answers to all five tasks. It is 17 pages long. Two pages fit on each screen page.

Read the whole script first. Then **fill in** grid 1 on your worksheet.

Candidate 1 evidence, National 5 Engineering Science Assignment 2024, Understanding Standards. © Qualifications Scotland (SQA).

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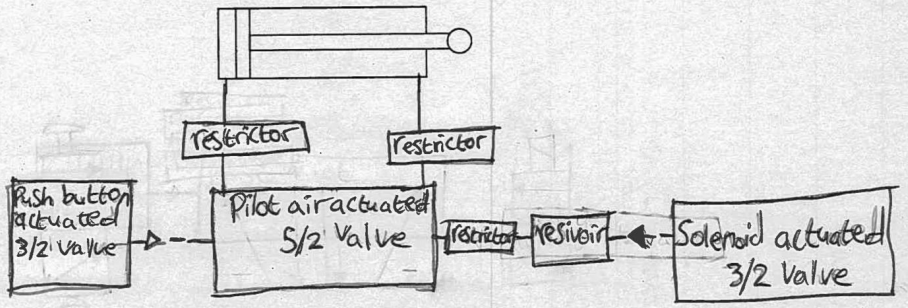
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Worksheet 1a

Actuator	Solenoid
Justification	Can create a microcontrolled circuit for the actuator that senses when a train passes using said microcontroller. This would be best because it would be easier due to there just being a microcontroller code, therefore much simpler to activate.

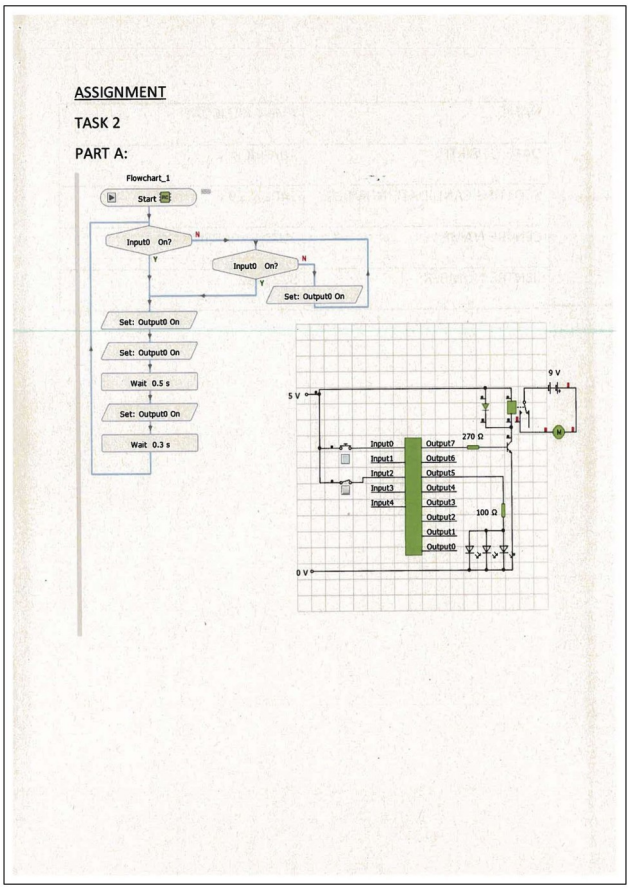
(2 marks)

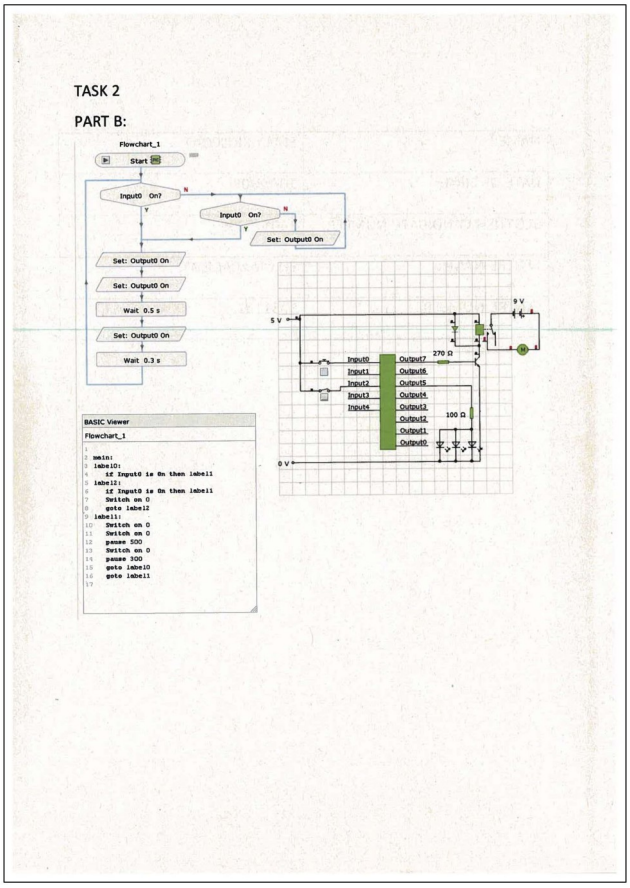
Worksheet 1b



(5 marks)

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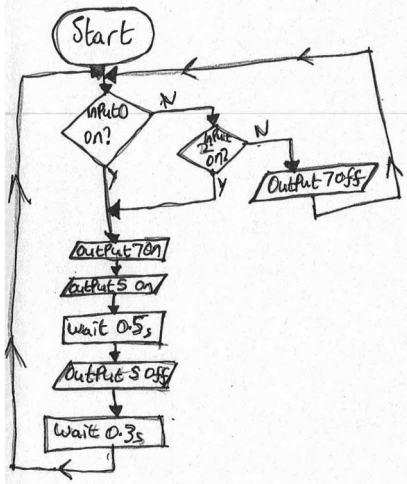
Worksheet 2c

Planned test	Expected result	Initial test result	Planned amendment to circuit/flowchart	Retest result
Test 1 Turn on the test switch **(input 2)** for around 20 seconds and then turn it off.	The motor starts to turn and the LEDs flash. When the test switch is turned off, the LEDs turn off and the motor slows to a stop.	*When the test switch is turned on ...* ~~Nothing happened~~ Transistor + Diode exploded, LEDS flashed on + off *When the test switch is turned off ...* ~~Nothing happened~~ LEDS turn off	~~Flowchart amendment (see other sheet)~~ Diode flipped around	*When the test switch is turned on ...* LED's flash on + off Motor spins *When the test switch is turned off ...* LED's turn off Motor slows to a stop
Planned test	**Expected result**	**Initial test result**	**Planned amendment to circuit/flowchart**	**Retest result**
Test 2 Press and hold the movement sensor **(input 0)** for around 20 seconds and then release it.	The motor starts to turn and the LEDs flash. When the movement sensor is released, the LEDs turn off and the motor slows to a stop.	*When the movement sensor is pressed ...* Nothing happens *When the movement sensor is released ...* Nothing happens.	Flowchart amendment (see on other sheet)	*When the movement sensor is pressed ...* LED's flash on + off Motor spins *When the movement sensor is released ...* LED's turn off Motor slows to a stop

(6 marks)

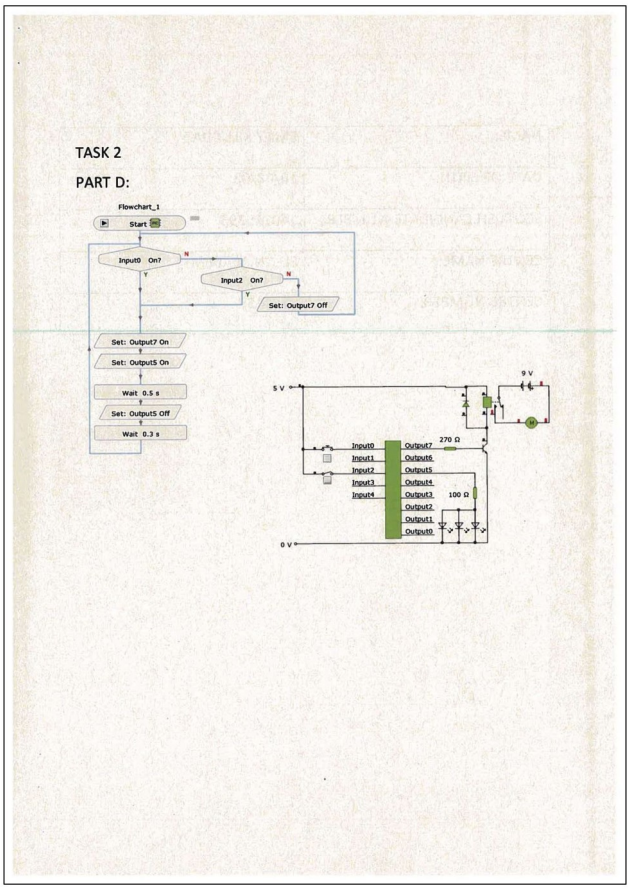
Task 2

Part C - Ammended Flowchart

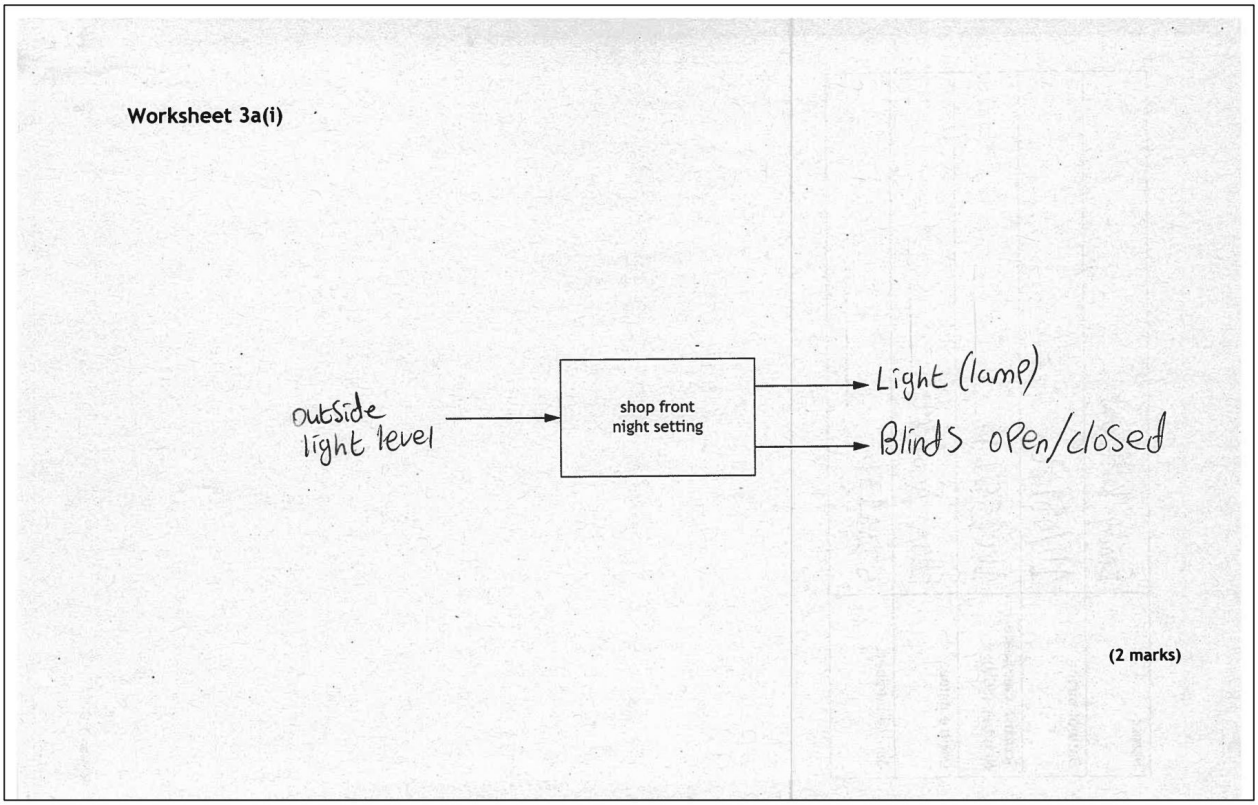


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Script 1 · page 7 of 17

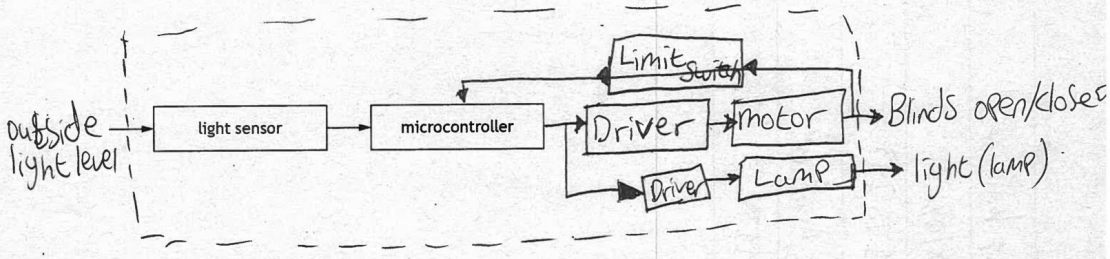


Script 1 · page 8 of 17



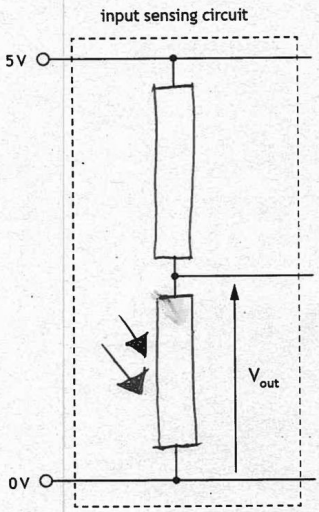
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Worksheet 3a(ii)



(6 marks)

Worksheet 3b



(3 marks)

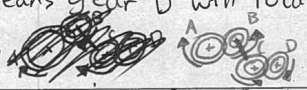
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Worksheet 3c

Planned test	Expected result
Test 1 Reduce the light level.	The lamp will turn on and the motor will fully close the blind
Planned test	**Expected result**
Test 2 Increase the light level	The lamp will turn off and the motor will open the blind fully.

(3 marks)

Worksheet 4a

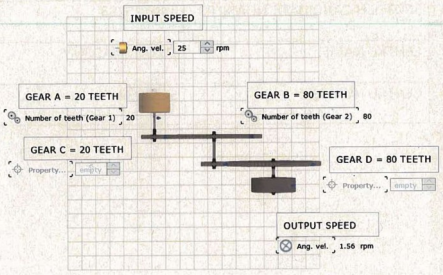
Specification point	Met? Yes/No	Justification
i	Yes	The design is a compound gear train, which means it will fit into a compact space inside the rotating display stand.
ii	Yes	Gears B and C ~~act as idler gear~~ will rotate in the same way, which means gear D will rotate in the same direction as gear A: 
iii	NO	MMR = $\frac{\text{driven}}{\text{driver}} \times \frac{\text{driven}}{\text{driver}} = \frac{15}{60} \times \frac{16}{16} = \frac{1}{4} \times 1 = 1/4 = 1:4$ VR: $\frac{\text{input}}{\text{output}} = \frac{25}{100} = 1/4 = 1:4$ AB: input speed × input size = output speed × output size 25 × 60 = output speed × 15 output speed = 1500/15 = 100 rev min^{-1} CD: input speed × input size = output speed × output size 100 × 16 = output speed × 16 output speed = 1600/16 = 100 rev min^{-1} This doesn't meet requirements because it increases the speed, it doesn't reduce it

(3 marks)

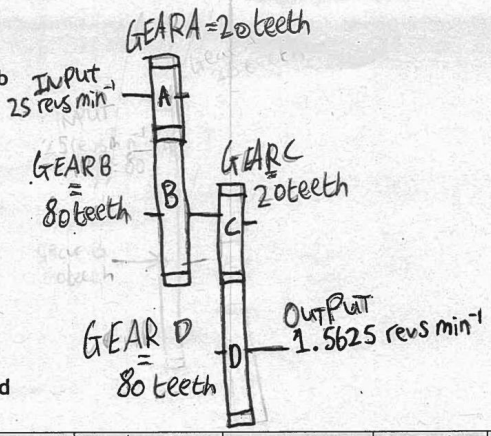
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TASK 4

PART C:



Worksheet 4b



(2 marks)

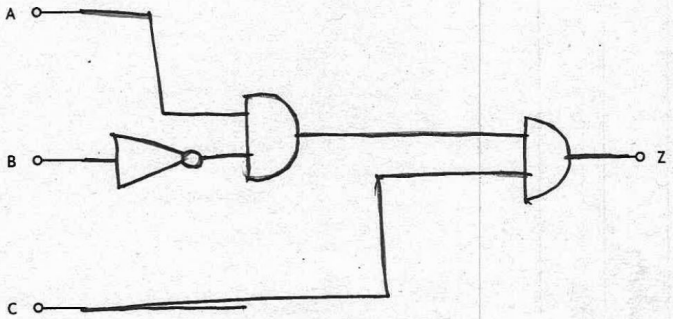
Worksheet 4d

Planned test	Input speed	Output speed	Required Velocity Ratio	Actual Velocity Ratio
Measure the input speed and output speed of the gear system and calculate the actual velocity ratio.	25 revs min⁻¹	1.56 rev min⁻¹	Between 12:1 and 20:1	16.03:1

(2 marks)

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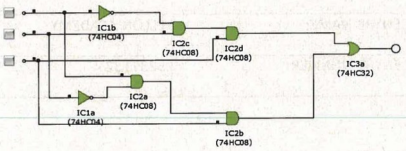
Worksheet 5a



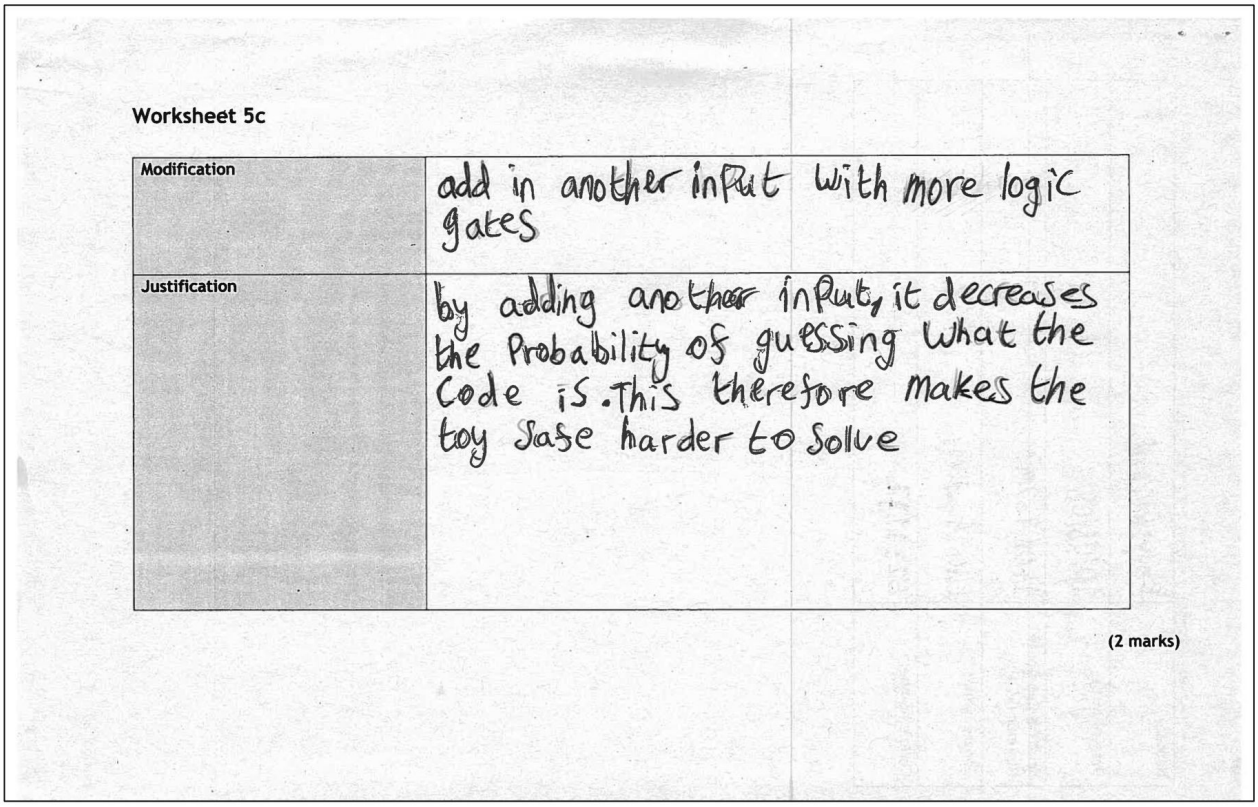
(2 marks)

TASK 5

PART B:



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Worksheet 5c

Modification	add in another input with more logic gates
Justification	by adding another input, it decreases the Probability of guessing what the code is. This therefore makes the toy case harder to solve

(2 marks)

End of script 1. **Fill in** grid 1 on your worksheet now.

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6 Script 2

Candidate 2's answers to all five tasks. It is 16 pages long. Two pages fit on each screen page.

Read the whole script first. Then **fill in** grid 2 on your worksheet.

Candidate 2 evidence, National 5 Engineering Science Assignment 2024, Understanding Standards. © Qualifications Scotland (SQA).

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Task 1A:

1a Choose a suitable actuator from the table above, to sense when a train has passed.

Complete worksheet 1a, justifying your choice.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(2 marks)

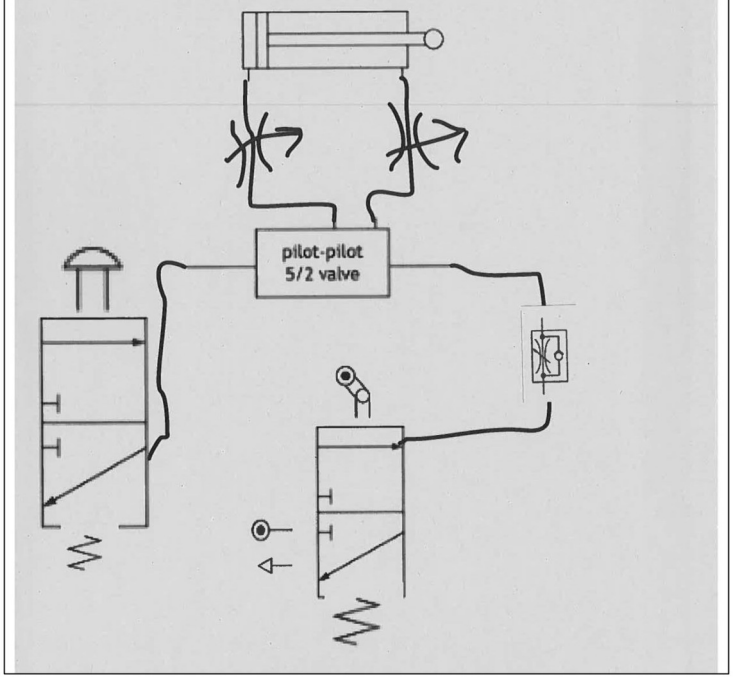
Actuator	Roller trim
Justification	Because it will detect when the train passes.

Task 1B:

Task 1 – model railway (continued)

1b Complete the design for the pneumatic circuit on **worksheet 1b**, to meet the specification, using your selected actuator from **task 1a**.

You must identify the name of each component, valve, and actuator, and indicate the direction of the piston outstroke. Show connections between all components and valves.



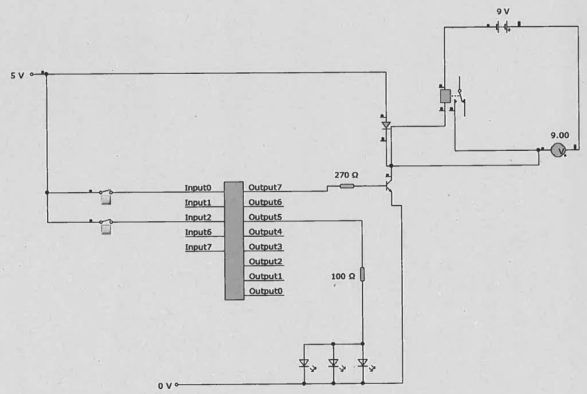
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Task 2a:

2a Simulate the flowchart sequence and electronic circuit **integrated together**, as shown on the following page. You can use a different sized microcontroller but the pin numbers must match the connections given.

Note: for test purposes, the movement sensor is replaced with a push-to-make switch.

Present your evidence on A4 single-sided pages, with the task number clearly labelled. Screenshots or images must be clear and easy to read.



Task 2b:

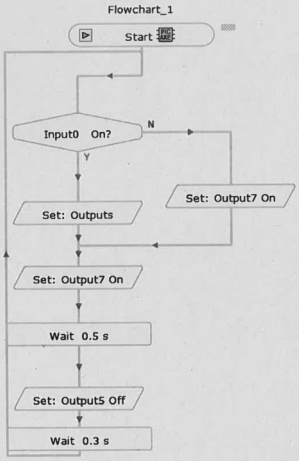
2b Produce high-level microcontroller code to fully match the flowchart from **task 2a.**

Present your evidence on A4 single-sided pages, with the task number clearly labelled. Screenshots or images must be clear and easy to read.

The microcontroller must meet the following specification:

i Whenever the test switch (input 2) is activated, the motor will rotate, and the LEDs will flash on and off continuously.
ii Whenever the movement sensor (input 0) detects a customer, the motor will rotate, and the LEDs will flash on and off continuously.
iii When the movement sensor no longer detects a customer, the motor and the LEDs will switch off.

Errors were found in both the flowchart and electronic circuit during testing.



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Task2c:

2c Complete the testing table on **worksheet 2c**, by carrying out the planned tests and describing each initial test result.

Describe your planned amendments to the flowchart and/or circuit, so that it meets the specification and make the amendments.

Carry out each test again, describing the result of your retest, before moving onto the next test.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

(6 marks)

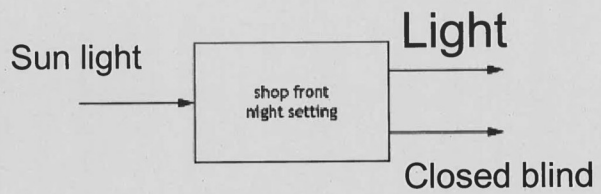
Planned test	Expected result	Initial test result	Planned amendment to circuit/flowchart	Retest result
Test 1 Turn on the test switch (input 2) for around 20 seconds and then turn it off.	The motor starts to turn and the LEDs flash. When the test switch is turned off, the LEDs turn off and the motor slows to a stop.	*When the test switch is turned on ...* **The transistor and diode blow up the LEDs flash and the motor turns** *When the test switch is turned off ...* **The motor stops slowly and LEDs turn off**	**Flip the diode the other way around**	*When the test switch is turned on ...* **The motor starts to turn and the led flash** *When the test switch is turned off ...* **The LEDs turn off and the motor turns off slowly.**
Planned test	**Expected result**	**Initial test result**	**Planned amendment to circuit/flowchart**	**Retest result**
Test 2 Press and hold the movement sensor (input 0) for around 20 seconds and then release it.	The motor starts to turn and the LEDs flash. When the movement sensor is released, the LEDs turn off and the motor slows to a stop.	*When the movement sensor is pressed ...* **The transistor and diode blow up the LEDs flash and the motor turns** *When the movement sensor is released ...* **The motor stops slowly and LEDs turn off**	**Flip the diode the other way around**	*When the movement sensor is pressed ...* **The motor starts to turn and the led flash** *When the movement sensor is released ...* **The LEDs turn off and the motor turns off slowly.**

Task 3a:

3a(i) Complete the system diagram on **worksheet 3a(i)** to meet the specification.

You must clearly show all external inputs and outputs.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.



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Task3a:

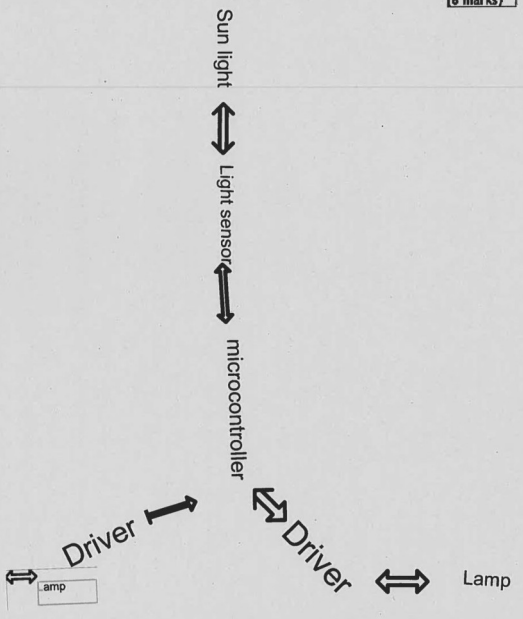
ii) Complete the sub-system diagram on **worksheet 3a(ii)** to meet the specification.

You must clearly show all external inputs and outputs, all sub-systems, the system boundary and the interactions between sub-systems.

Note: each output device requires a driver.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

[6 marks]

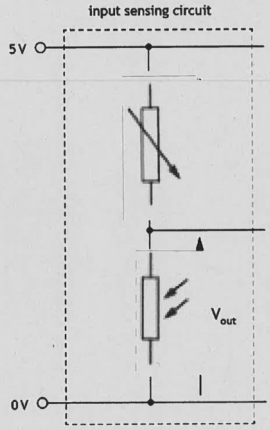


Task 3b:

3b Complete the circuit diagram on **worksheet 3b**, by designing a suitable input sensing circuit that will produce a high signal (V_{out}) when it is dark. Component values are not required.

Note: you must not construct or use simulation software to complete this part of the task.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.



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Task3c:

3c Using the input sensor circuit you designed in task 3b, complete the test plan on **worksheet 3c** by describing:

- the expected result from the planned test given in terms of V_{out}
- a second planned test and the expected result in terms of V_{out}

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

Planned test	Expected result
Test 1 Reduce the light level.	**Output 1: light turns on and blinds close**

Planned test	Expected result
Test 2 **Increase light level**	**Output 2: light turns off and opens blind**

Task4a :

4a Evaluate the initial gear train design against the specification and complete the table on **worksheet 4a**. You should:

- state if each specification point is met
- justify your decision

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

Specification point	Met? Yes/No	Justification
i	no	**Gear A is is too big to fit in a tight space**
ii	yes	**Gear B and C act as an idler which makes gear D spin in the same direction as gear A**
iii	no	**The output speed is significantly greater than the input speed**

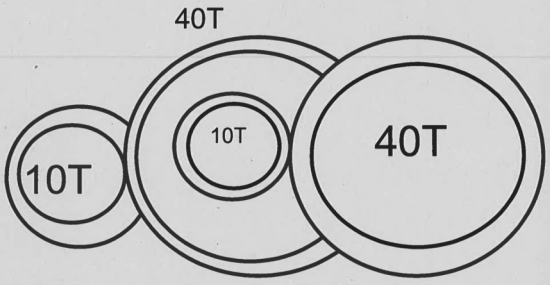
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Task4b:

4b Design a gear train on **worksheet 4b**, based on the gear sizes available to you. This needs to give a speed reduction of at least a factor of 12, but not greater than a factor of 20. Label the input gear and include all gear sizes.

Note: you must not construct or use simulation software to complete this part of the task.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

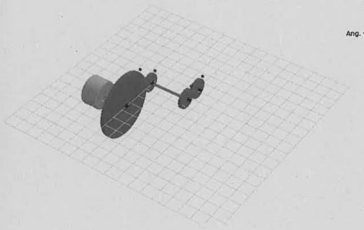


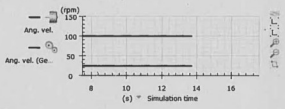
Out/in= 10/40x 10/40
Out/20= 1/16
Out = 1.25 RPM

Task 4c:

4c Simulate or construct your design from **task 4b**. You must include an input component to allow for testing and label all gear sizes.

Present your evidence on A4 single-sided pages, with the task number clearly labelled. Screenshots or images must be clear and easy to read.





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Task 4d:

4d Complete the test result table on worksheet 4d by:

- entering the results from task 4c that you observed during simulation or after construction
- calculating the actual velocity ratio, using your recorded values for the input and output speeds

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

Planned test	Input speed	Output speed	Required Velocity Ratio	Actual Velocity Ratio
Measure the input speed and output speed of the gear system and calculate the actual velocity ratio.	20 RPM	1.25RPM	Between 12:1 and 20:1	16:1

VR= in/out

VR= 20/1.25 RPM

VR= 16:1

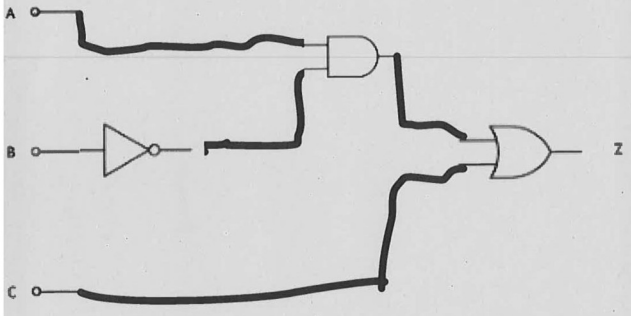
Task 5a:

5a Complete the logic diagram on worksheet 5a to unlock the safe by sending a high (1) signal to output Z when:

input A is high (1) and input B is low (0) and input C is high (1)

Note: you must not construct or use simulation software to complete this part of the task.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.

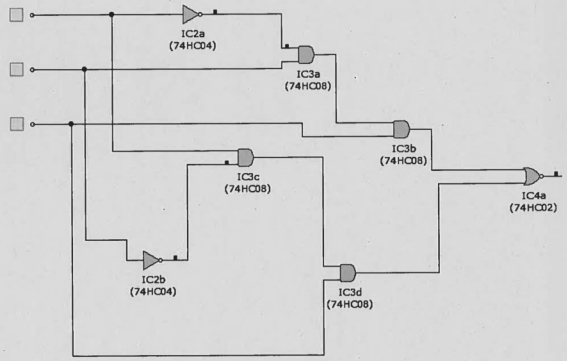


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Task 5b:

5b Simulate or construct the logic diagram shown above. You must include suitable input and output devices to allow for potential testing. If constructed, you must clearly label the integrated circuit (IC) numbers.

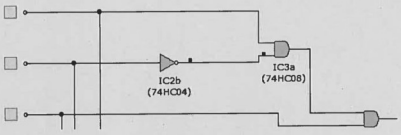
Present your evidence on A4 single-sided pages, with the task number clearly labelled. Screenshots or images must be clear and easy to read.



Task5c :

5c Using worksheet 5c, describe a suitable modification to the logic circuit in task 5b that would make the code for the toy safe harder to solve and justify your reason.

Present your evidence on A4 single-sided pages, with the task number clearly labelled.



Modification	**Move the not gate from the A input to the B input**
Justification	**Previously the only time when the outputs was high was when ABC were high and when the AC were high and now. With the not gate moved the only correct combination is AC**

End of script 2. **Fill in** grid 2 on your worksheet now.

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7 Script 3

Candidate 3's answers to all five tasks. It is 16 pages long. Two pages fit on each screen page.

Read the whole script first. Then **fill in** grid 3 on your worksheet.

Candidate 3 evidence, National 5 Engineering Science Assignment 2024, Understanding Standards. © Qualifications Scotland (SQA).

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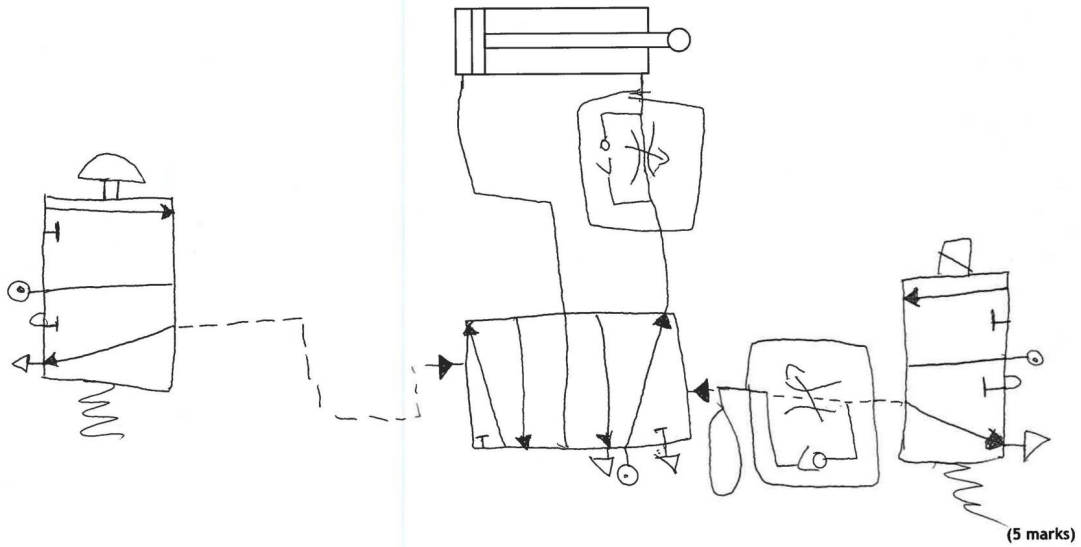
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Worksheet 1a

Actuator	Electrical solenoid
Justification	This was my choice because there will be no person needed to switch any switches and there won't be any contact needed when switching between rails. It will be easier and more efficient for train companies.

(2 marks)

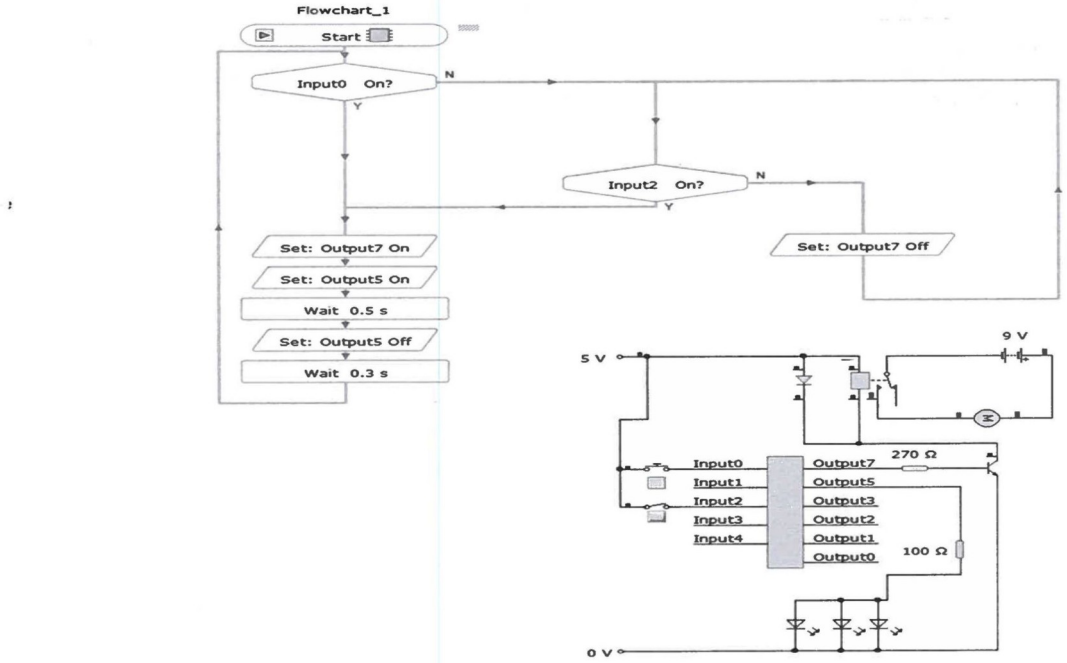
Worksheet 1b



(5 marks)

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Task 2a



Task 2b

BASIC Viewer

Flowchart_1

```
1
2 main:
3 label0:
4    if Input0 is On then label1
5 label2:
6    if Input2 is On then label1
7    Switch off 5
8    goto label2
9 label1:
10   Switch on 5
11   Switch on 4
12   pause 500
13   Switch off 4
14   pause 300
15   goto label0
16   goto label1
17
```

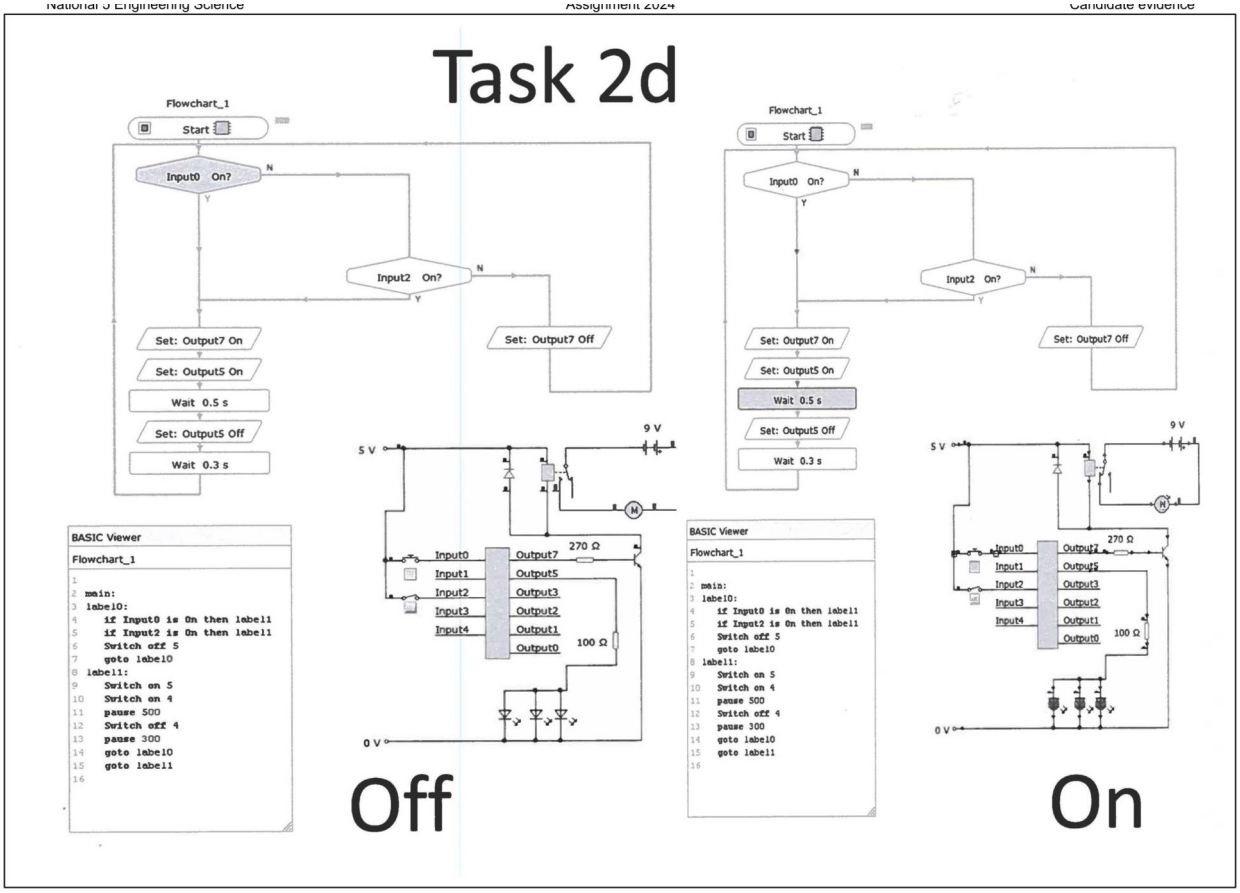

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Worksheet 2c

Planned test	Expected result	Initial test result	Planned amendment to circuit/flowchart	Retest result
Test 1 Turn on the test switch **(input 2)** for around 20 seconds and then turn it off.	The motor starts to turn and the LEDs flash. When the test switch is turned off, the LEDs turn off and the motor slows to a stop.	*When the test switch is turned on ...* The diode and the NPN transistor blow up. The motor stops and the LEDs turn on and then off. *When the test switch is turned off ...* The motor stays stopped and the LEDs don't turn back on.	I will move the diode 180° so it is on its correct side because it was upside down.	*When the test switch is turned on ...* The LEDs turn on and off repeatedly and the motor turns *When the test switch is turned off ...* The LEDs stop and the motor slowly comes to an end.
Planned test	**Expected result**	**Initial test result**	**Planned amendment to circuit/flowchart**	**Retest result**
Test 2 Press and hold the movement sensor **(input 0)** for around 20 seconds and then release it.	The motor starts to turn and the LEDs flash. When the movement sensor is released, the LEDs turn off and the motor slows to a stop.	*When the movement sensor is pressed ...* Nothing happens it doesn't turn on. *When the movement sensor is released ...* Nothing happens it stays the way it was.	I made it an or Programme by changing the input 7 off so it will go back to the start and re-ask the question is input '0' on?	*When the movement sensor is pressed ...* The motor starts and the LED's go on and off *When the movement sensor is released ...* The motor slowly comes to a stop and the LED's stop flashing.

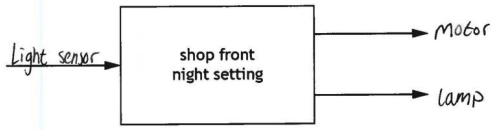
(6 marks)

Task 2d



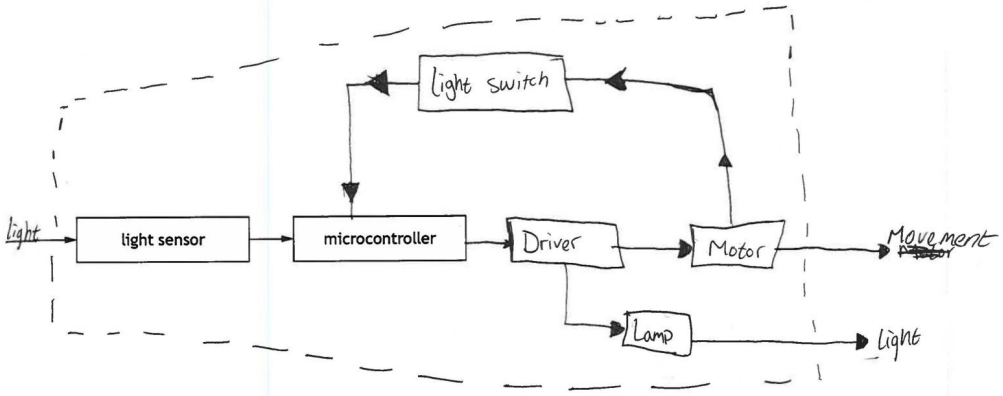
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Worksheet 3a(i)



(2 marks)

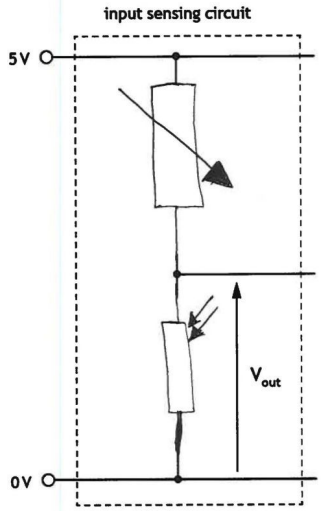
Worksheet 3a(ii)



(6 marks)

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Worksheet 3b



(3 marks)

Worksheet 3c

Planned test	Expected result
Test 1 Reduce the light level.	The light dependent resistor will sense lower level of light, because of this the resistance will be increased helping it produce a high signal.
Planned test	**Expected result**
Test 2 change the resistance of the variable resistor	This will allow the light dependent resistor (LDR) to be adjusted to allow it to sense a lighter or darker light level.

(3 marks)

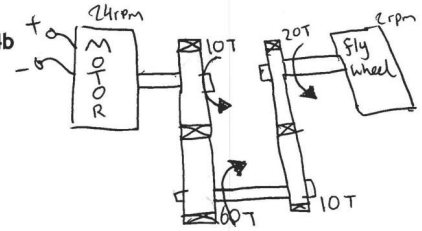
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Worksheet 4a

Specification point	Met? Yes/No	Justification
i	NO	because they are all spaced out / the gaps in between them are too big, meaning it will not fit into compact space.
ii	Yes.	because gear 'A' spins clockwise - this will force gear 'B' to spin anticlockwise because 'B' and 'C' are connected together / joined together they will spin in the same way. because 'C' is going anticlockwise it will force 'D' to spin clockwise.
iii	NO.	because the gear chain starts with a big gear instead of a small gear because of this the gears will make it that the output speed will be bigger than 9.

(3 marks)

Worksheet 4b



$VR = \frac{Dn}{Dr} \times \frac{Dn}{Dr}$

$= \frac{60}{10} \times \frac{20}{10}$

$= \frac{6}{1} \times \frac{2}{1}$

$= 12$

(2 marks)

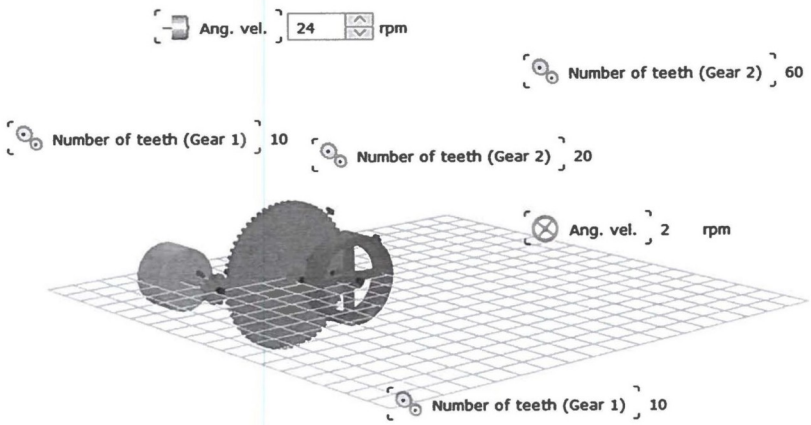
Worksheet 4d

Planned test	Input speed	Output speed	Required Velocity Ratio	Actual Velocity Ratio
Measure the input speed and output speed of the gear system and calculate the actual velocity ratio.	24 rpm	2rpm	Between 12:1 and 20:1	$\frac{60}{10} \times \frac{20}{10}$ $= \frac{1200}{100}$ $= 12$

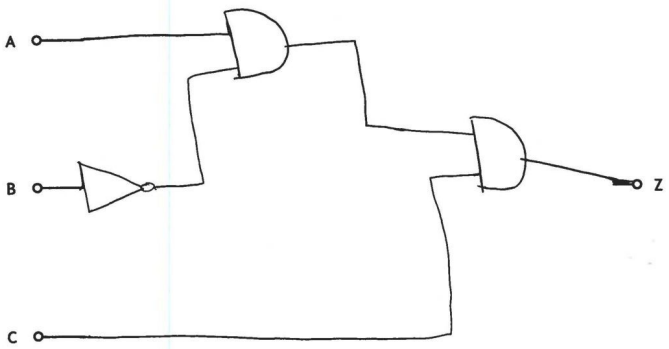
(2 marks)

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Task 4c



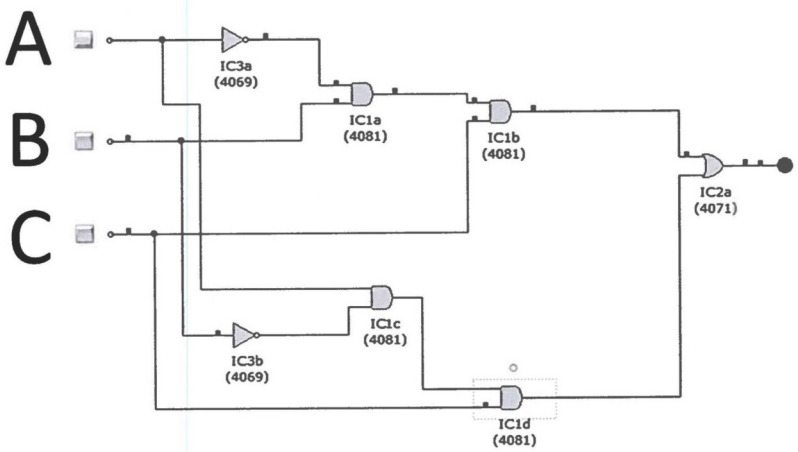
Worksheet 5a



(2 marks)

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Task 5b



Worksheet 5c

Modification	I would make it have more inputs e.g A B C D E
Justification	This will make it that the person will have to guess more numbers and a lot more number combinations. This will make it harder and challenging so no-one will break into your safe.

(2 marks)

End of script 3. **Fill in** grid 3 on your worksheet now.

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8 Answers

These are the marker's marks, from the Understanding Standards commentary. **Mark the scripts yourself before you look.**

Task	What it asks for	Max	Script 1	Script 2	Script 3
1a	Actuator and justification	2	2	2	1
1b	Pneumatic circuit	5	4	3	4
2a	Flowchart and circuit	5	3	1	4
2b	High-level code	1	1	0	0
2c	Test results and amendments	6	4	1	4
2d	Amendments carried out	2	2	0	2
3a(i)	System diagram: input and outputs	2	2	2	0
3a(ii)	Sub-system diagram	6	4	2	3
3b	Input sensing circuit	3	3	2	3
3c	Test plan: Vout	3	1	1	1
4a	Evaluate the gear design	3	3	3	3
4b	Design a gear train	2	2	1	2
4c	Simulate the gear train	2	2	1	2
4d	Velocity ratio	2	2	1	2
5a	Logic diagram	2	2	1	2
5b	Simulate the logic	2	2	1	2
5c	Modify and justify	2	2	1	2
	Total	**50**	**41**	**23**	**37**

Focus task answers

Task 1 (2c). Question 1, ticks for scripts 1, 2, 3. Test 1 result: no, no, yes. Diode change: yes, yes, yes. Test 1 re-test: yes, no, yes. Test 2 result: no, no, no. Loop-back change: yes, no, no. Test 2 re-test: yes, no, yes. Question 2: script 1 says what the LEDs do but not what the motor does. Question 3: when the test switch is turned on, the LEDs flash and the motor does not turn.

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Task 2 (3a(ii)). Question 1, ticks for scripts 1, 2, 3. Boundary and boxes: yes, no, yes. Motor as an output: yes, no, yes. Lamp as an output: yes, yes, yes. One driver per output: no, yes, no. Two limit switches: no, no, no. Feedback loop: yes, no, no. Question 2: script 3 used one driver for both outputs. Each output needs its own driver, joined to the microcontroller. Question 3: the loop starts after the motor, goes through a limit switch, and goes into the microcontroller, with both arrowheads.
Task 3 (1b). Question 1, ticks for scripts 1, 2, 3. Cylinder to 5/2 valve: yes, yes, yes. Two restrictors: no, no, no. Both 3/2 valves to the 5/2 valve: yes, yes, yes. Push button and actuator on the right valves: yes, yes, yes. Reservoir: yes, no, yes. Question 2: the mark needs two uni-directional restrictors, so that both the outstroke and the instroke are slowed. No script restricted the air in both pipes. Question 3: after the instroking 3/2 valve, on the line to the 5/2 valve. Script 2 left it out.
Task 4 (3c). Question 1: every script gets one tick only, for a suitable second test. No script says what happens to Vout. Question 2: script 1 describes the lamp and the blind. The mark needs the effect on Vout. Question 3: Test 1, reduce the light level: Vout will increase. Test 2, increase the light level: Vout will decrease.
Task 5 (5c). Question 1, ticks for scripts 1, 2, 3. Valid change: yes, yes, yes. Clear justification: yes, no, yes. Question 2: script 3 says there are a lot more number combinations to guess. "Harder" on its own does not score. Question 3: adding another input increases the number of possible combinations for the code.
Task 6 (3a(i)). Question 1, ticks for scripts 1, 2, 3. Input is a light level: yes, yes, no. Outputs are light and movement: yes, yes, no. Question 2: script 3 names components (a light sensor, a motor and a lamp). The marking instructions do not accept components. Question 3: input: outside light level. Outputs: light, and movement (the blind opens or closes).

The full commentary follows. Read the marker's reasons for each task.

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Commentary on candidate evidence

The candidate evidence has achieved the following marks for each task in this assignment.

Candidate 1

Task 1a

The candidate was awarded **2 marks**:

- for selecting an appropriate actuator – solenoid (**1 mark**)
- for justification in context of sensing train (**1 mark**)

Task 1b

The candidate was awarded **4 marks**:

- two pipes between cylinder and 5/2 valve (**1 mark**)
- not only restricting just exhaust air (**0 mark**)
- 3/2 valves connected to the 5/2 valve (**1 mark**)
- the push button and selected actuator on outstroking and instroking 3/2 valves respectively (**1 mark**)
- reservoir on instroking air to 5/2 valve (**1 mark**)

Task 2a

The candidate was awarded **3 marks**:

- all stages and symbols were noted in the flowchart **(1 mark)**
- flowchart pin numbers error with second decision and all output boxes (**0 mark)**
- selection of correct circuit components (**1 mark**)
- circuit wiring – incorrect battery orientation (**0 mark**)
- correct integration of flowchart and circuit (**1 mark**)

Task 2b

The candidate was awarded **1 mark** because the coding matched the flowchart in task 2a.

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Task 2c

The candidate was awarded **4 marks**:

- initial test result had no comment on the effect on the motor (**0 mark**)
- planned amendment – diode flipped (**1 mark**)
- correct retest results based on circuit in 2d (**1 mark**)
- initial retest result has no statement on the LED and motor (**0 mark**)
- planned amendment described graphically in the flowchart sketch (**1 mark**)
- correct retest result for both tests (**1 mark**)

Task 2d

The candidate was awarded **2 marks**:

- for carrying out the first planned amendment – diode flipped (**1 mark**)
- for completing the second planned amendment – looping to the start of the flowchart (**1 mark**)

Task 3a(i)

The candidate was awarded **2 marks**:

- for identifying the correct input – outside light (**1 mark**)
- as both outputs are correctly labelled – light and movement (inferred by blinds open/closed) (**1 mark**)

Task 3a(ii)

The candidate was awarded **4 marks**:

- system boundary and boxes around all sub-systems (**1 mark**)
- motor in the output position (**1 mark**)
- lamp in the output position (**1 mark**)
- a driver per output – these are not individually joined to the microcontroller (**0 mark**)
- both limit switches are individually connected to microcontroller (**0 mark**)
- feedback loop after motor into the microcontroller via a limit switch with both arrows (**1 mark**)

Task 3b

The candidate was awarded **3 marks**:

- for the LDR symbol (**1 mark**)
- for the resistor symbol (**1 mark**)

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- connecting components as a dark sensor (**1 mark**)

Task 3c

The candidate was awarded **1 mark**:

- there is no description of the effect on V_{out} when the light level is reduced (**0 mark**)
- for stating an appropriate planned test of the circuit in 3b – increase the light level (**1 mark**)
- no description was given for the effect on V_{out} caused by the candidate's planned test (**0 mark**)

Task 4a

The candidate was awarded **3 marks**:

- for justifying specification point i. met by referring to the use of a compound gear train (**1 mark**)
- for justifying the meeting of specification point ii. by referencing the gear directions (**1 mark**)
- for justifying that specification point iii. was not met through calculations to show a speed increase (**1 mark**)

Task 4b

The candidate was awarded **2 marks**:

- for a gear train with all gear sizes shown (**1 mark**)
- for producing a speed reduction of 16:1 (**1 mark**)

Task 4c

The candidate was awarded **2 marks**:

- for simulating the gear type shown in task 4b (**1 mark**)
- for simulating with the gear sizes in task 4b (**1 mark**)

Task 4d

The candidate was awarded **2 marks**:

- for the input and output speeds from task 4c (**1 mark**)
- for correct velocity ratio using input/output speeds (**1 mark**)

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Task 5a

The candidate was awarded **2 marks**:

- for inverting input B (**1 mark**)
- for ANDing inputs A, B, and C (**1 mark**)

Task 5b

The candidate was awarded **2 marks**:

- $\overline{A} \cdot B \cdot C$ logic gate connections (**1 mark**)
- $A \cdot \overline{B} \cdot C$ logic gate connections (**1 mark**)

Task 5c

The candidate was awarded **2 marks**:

- for a valid modification suggested to the logic circuit (**1 mark**)
- for the justification of the suggested modification to the logic circuit (**1 mark**)

Total marks awarded – 41/50

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Candidate 2

Task 1a

The candidate was awarded **2 marks**:

- for selecting an appropriate actuator – roller trim, the spelling was not penalised (**1 mark**)
- for justification of actuator in the context of detecting a train (**1 mark**)

Task 1b

The candidate was awarded **3 marks**:

- two pipes between the cylinder and the 5/2 valve (**1 mark**)
- not only restricting just exhaust air (**0 mark**)
- connecting both 3/2 valves to the 5/2 valve (**1 mark**)
- the push button and roller trip actuator shown on the correct 3/2 valve for the movement (**1 mark**)
- no reservoir on the instroking air to the 5/2 valve (**0 mark**)

Task 2a

The candidate was awarded **1 mark**:

- in the flowchart pin numbers are incorrect with 'set: outputs' command and no output 5 on (**0 mark**)
- there is no motor in any of the circuit components (**0 mark**)
- circuit wiring incorrect due to incorrect battery orientation (**0 mark**)
- integration of the flowchart and circuit (**1 mark**)

Task 2b

The candidate was awarded **0 marks** because they did not provide a high level code for the flowchart.

Task 2c

The candidate was awarded **1 mark**:

- the initial test result has no motor in the circuit (**0 mark**)
- for planned amendment – flipping diode the other way around (**1 mark**)
- retest results have no statement about adding a motor or evidence presented for task 2d (**0 mark**)
- in the second initial retest result there is no motor in task 2a, so this statement is incorrect (**0 mark**)

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- the second planned amendment is incorrect and will return the diode to its original incorrect orientation (**0 mark**)
- the retest results statements are incorrect for evidence in task 2a (**0 mark**)

Task 2d

The candidate was awarded **0 marks** as they did not provide a modification to flowchart.

Task 3a(i)

The candidate was awarded **2 marks**:

- for identifying the correct input – sunlight (**1 mark**)
- both outputs are correctly labelled – light and movement (inferred by closed blind) (**1 mark**)

Task 3a(ii)

The candidate was awarded **2 marks**:

- no system boundary or boxes around all sub-systems (**0 mark**)
- no motor included (**0 mark**)
- the lamp(s) in an output position (**1 mark**)
- a driver per output as each lamp has a driver from the microcontroller (**1 mark**)
- no limit switch connected to the microcontroller (**0 mark**)
- lack of a correct feedback loop (**0 mark**)

Task 3b

The candidate was awarded **2 marks**:

- for the LDR symbol (**1 mark**)
- for the (variable) resistor symbol (**1 mark**)
- connecting components as a dark sensor because of the wiring is incomplete (**0 mark**)

Task 3c

The candidate was awarded **1 mark**:

- the description does not reference any effect on V_{out} by the reduction in the light level (**0 mark**)
- for an appropriate planned test – increase the light level (**1 mark**)

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- there is no description of the effect on V_{out} caused by the candidate's planned test (**0 mark**)

Task 4a

The candidate was awarded **3 marks**:

- for justifying that specification point i. is not met by referring to gear A (60 tooth) being too big (**1 mark**)
- for justifying the satisfying of specification point ii. with reference to gear directions (**1 mark**)
- for justifying that specification point iii. was not met with reference to an increase in speed being produced (**1 mark**)

Task 4b

The candidate was awarded **1 mark**:

- for the gear train with all gear sizes shown (**1 mark**)
- due to the increase rather than reduction in output speed (**0 mark**)

Task 4c

The candidate was awarded **1 mark**:

- for simulating gear train type shown in task 4b (**1 mark**)
- the gear sizes do not match those in task 4b, evidenced from the speed graphs values (**0 mark**)

Task 4d

The candidate was awarded **1 mark**:

- input and output speeds from task 4c have not been used (**0 mark**)
- for velocity ratio based on given input/output speeds (**1 mark**)

Task 5a

The candidate was awarded **1 mark**:

- for inverting input B (**1 mark**)
- for using ANDing inputs A, B, and C as an OR gate (**0 mark**)

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Task 5b

The candidate was awarded **1 mark**:

- $\overline{A} \cdot B \cdot C$ logic gate connections (**1 mark**)
- $A \cdot \overline{B} \cdot C$ logic gate connections as a NOR rather than an OR was used at the output (**0 mark**)

Task 5c

The candidate was awarded **1 mark**:

- for valid modification to logic circuit (**1 mark**)
- the justification statement is unclear (**0 mark**)

Total marks awarded – 23/50

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Candidate 3

Task 1a

The candidate was awarded **1 mark**:

- for selecting a suitable actuator – solenoid (**1 mark**)
- justification of sensing a train was not mentioned (**0 mark**)

Task 1b

The candidate was awarded **4 marks**:

- two pipes between the cylinder and 5/2 valve (**1 mark**)
- not restricting exhaust air in both pipes (**0 mark**)
- both 3/2 valves connected to the 5/2 valve (**1 mark**)
- the push button and solenoid actuator on the correct 3/2 valve for the required movement (**1 mark**)
- the reservoir on the instroke 5/2 valve actuator line (**1 mark**)

Task 2a

The candidate was awarded **4 marks**:

- for all flowchart stages with the correct symbols (**1 mark**)
- for flowchart pins, decisions, and time delays (**1 mark**)
- for selecting all correct circuit components (**1 mark**)
- the circuit wiring due to battery orientation (**0 mark**)
- for integration of the flowchart and circuit (**1 mark**)

Task 2b

The candidate was awarded **0 marks** as the pin numbers do not match with task 2a.

Task 2c

The candidate was awarded **4 marks**:

- the initial test result for task 2a with the motor off and flashing LEDs (**1 mark**)
- for planned amendment – move diode 180^0 (**1 mark**)
- correct retest results from task 2d changes (**1 mark**)
- in the second initial retest result – nothing happens, as no motor and LED comment is included (**0 mark**)
- there is a second planned amendment error – loop back from input 7, this should have stated output 7 (**0 mark**)

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- for correct retest results for evidence in 2d (**1 mark**)

Task 2d

The candidate was awarded **2 marks**:

- for carrying out first planned amendment – move diode 180^0 (**1 mark**)
- for completing second planned amendment – looping to start of the flowchart (**1 mark**)

Task 3a(i)

The candidate was awarded **0 marks**:

- incorrect external input – light sensor (**0 mark**)
- incorrect external outputs – motor and lamp (**0 mark**)

Task 3a(ii)

The candidate was awarded **3 marks**:

- for a system boundary and boxes around all the sub-systems (**1 mark**)
- for motor in an output position (**1 mark**)
- for lamp in an output position (**1 mark**)
- just one driver was used for both outputs (**0 mark**)
- no limit switches were connected to the microcontroller (**0 mark**)
- the feedback loop does not start after the motor or include a limit switch (**0 mark**)

Task 3b

The candidate was awarded **3 marks**:

- for the LDR symbol (**1 mark**)
- for the (variable) resistor symbol (**1 mark**)
- for wiring the components as a dark sensor (**1 mark**)

Task 3c

The candidate was awarded **1 mark**:

- the description does not reference any effect on V_{out} the when the light level is reduced (**0 mark**)
- for an appropriate planned test of the circuit in task 3b – change the resistance of the variable resistor (**1 mark**)

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- the description does not describe any effect on V_{out} of their planned test (**0 mark**)

Task 4a

The candidate was awarded **3 marks**:

- for justifying not meeting specification point i. by referring to gaps in the gear spacing (**1 mark**)
- for justifying meeting specification point ii. with reference to the gear direction of rotation (**1 mark**)
- for justifying not meeting specification point iii. by referring to increasing the output speed (**1 mark**)

Task 4b

The candidate was awarded **2 marks**:

- the gear train with all gear sizes shown (**1 mark**)
- for reducing the output speed by 12:1 (**1 mark**)

Task 4c

The candidate was awarded **2 marks**:

- for simulating the type of gear in task 4b (**1 mark**)
- for simulation gear sizes all matching 4b (**1 mark**)

Task 4d

The candidate was awarded **2 marks**:

- for correct input and output speeds from task 4c (**1 mark**)
- velocity ratio based on given input/output speeds (**1 mark**)

Task 5a

The candidate was awarded **2 marks**:

- for inverting input B (**1 mark**)
- for ANDing inputs A, B, and C (**1 mark**)

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Task 5b

The candidate was awarded **2 marks**:

- $\overline{A} \cdot B \cdot C$ logic connections (**1 mark**)
- $A \cdot \overline{B} \cdot C$ logic connections (**1 mark**)

Task 5c

The candidate was awarded **2 marks**:

- for adding more inputs (**1 mark**)
- for valid justification of the modification (**1 mark**)

Total marks awarded – 37/50

Credits: candidate evidence 1 to 3 and commentaries, National 5 Engineering Science Assignment 2024, Understanding Standards, © Qualifications Scotland (SQA). Assessment task (toy shop) and finalised marking instructions, National 5 Engineering Science Assignment 2024, © Qualifications Scotland (SQA); photographs in the task: Task 1 Galchenkova

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